LEDGER MVP

Underwriting Blueprint & Architecture

Dutch E-Commerce SME Working-Capital Lending

Credit Decision System · Local Python MVP · Demo-Ready

Document	Full Technical Blueprint
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Stack	Python 3.11+ · pandas · scikit-learn · DuckDB · Streamlit
Status	MVP / Demo Foundation — not production

1. Credit Decision Design

Decision Outputs

Every application produces a decision envelope with the following outputs. Per the business plan, Phase 1 uses a rules-based champion model reviewed by a credit officer for every loan. In Year 1, ALL loans go to manual review.

Output	Type	Values
decision	enum	APPROVE / DECLINE / MANUAL_REVIEW
max_amount	EUR int	0 – 150,000 (capped by policy)
pricing_band	enum	A (11%) / B (12.5%) / C (14%)
tenor_max_months	int	3 – 18
reason_codes	list[str]	Machine-readable codes
explanations	list[str]	Human-readable sentences
manual_review_flags	list[str]	Triggers for credit-officer sign-off
shadow_ml_score	float|None	0–1 probability (Phase 2, informational only)

Hard Knock-Out Rules (any fail → DECLINE)

#	Rule	Threshold	Reason Code
H1	Trading history	< 24 months	INSUFFICIENT_HISTORY
H2	KvK registration valid	Not active	KVK_INVALID
H3	Bank-PSP reconciliation	Mismatch > 20%	RECON_FAIL
H4	Negative balance frequency	> 50% of days (90d)	PERSISTENT_NEGATIVE_BALANCE
H5	Sanctions/PEP hit	Any	SANCTIONS_HIT
H6	Monthly GMV (trailing 6m)	< EUR 25,000	REVENUE_TOO_LOW

Scored Gates (weighted pass/fail)

#	Feature	Green (0 pts)	Amber (1 pt)	Red (2 pts)	Reason Code
S1	revenue_volatility_90d	CV < 0.25	0.25–0.45	> 0.45	HIGH_REV_VOLATILITY
S2	refund_rate_ltm	< 3%	3–8%	> 8%	HIGH_REFUND_RATE
S3	settlement_delay_p95	< 5 days	5–10 days	> 10 days	SETTLEMENT_DELAYS
S4	chargeback_rate_ltm	< 0.5%	0.5–1.5%	> 1.5%	HIGH_CHARGEBACKS
S5	platform_concentration	HHI < 0.40	0.40–0.65	> 0.65	PLATFORM_CONCENTRATION
S6	net_cashflow_coverage	> 1.8×	1.2–1.8×	< 1.2×	WEAK_CASHFLOW_COVERAGE
S7	supplier_pay_punctuality	> 90%	70–90%	< 70%	LATE_SUPPLIER_PAYMENTS
S8	ad_spend_ratio_3m	< 15%	15–30%	> 30%	HIGH_AD_DEPENDENCY

Scoring logic:

- Total red points >= 3 → DECLINE
- Total red points = 2 → MANUAL_REVIEW
- Total amber + red >= 5 → MANUAL_REVIEW

- Otherwise → APPROVE (still goes to credit officer in Year 1)

Max Amount Calculation

```
max_amount = min( requested_amount, avg_monthly_net_cashflow × tenor_months × 0.33,
avg_monthly_gmv × 2.5, EUR 25,000 if Year-1 pilot else EUR 150,000 )
```

Pricing Band Assignment

Band	APR	Criteria
A	11.0%	0 red, ≤ 1 amber, cashflow coverage > 2.0×
B	12.5%	Default / baseline
C	14.0%	1+ red (approved via manual review) or coverage < 1.5×

Shadow ML Model Design

Aspect	Specification
Target variable	default_30dpd — binary, 1 if loan reaches 30+ DPD (synthetic in demo)
Algorithm	GradientBoostingClassifier (scikit-learn); logistic regression as fallback
Features	Same 15-feature vector as champion policy
Output	shadow_score ∈ [0, 1] — P(default). No automated decisions.
Evaluation	Weekly comparison: for each champion decision, show ML score distribution
Governance	Shadow scores logged but never override policy. No ML superiority claims.

2. MVP Data Plan

A. PSD2 / Open Banking (Bank Transactions)

Field	Type	Example	Req
transaction_id	str	txn_abc123	✓
merchant_id	str (FK)	m_001	✓
date	date	2026-03-15	✓
amount	float	-1,250.00	✓
direction	enum	CREDIT / DEBIT	✓
counterparty_name	str	Mollie B.V.	✓
category	str	psp_settlement	✓
balance_after	float	14,320.50	✓

- **Join key:** merchant_id + iban
- **Reconciliation:** PSP settlement amounts in bank ↔ PSP payout records (tolerance ±2%)
- **Update cadence:** Daily (AISP pull) in production; static load in MVP
- **Consent:** Explicit PSD2 consent logged with timestamp, scope, revocation endpoint

B. PSP Data (Mollie / Adyen)

Field	Type	Example	Req
psp_transaction_id	str	psp_xyz789	✓
merchant_id	str (FK)	m_001	✓
psp_name	str	mollie	✓
order_date	date	2026-03-14	✓
settlement_date	date	2026-03-16	✓
gross_amount	float	89.99	✓
net_amount	float	88.78	✓
status	enum	paid / refunded / chargeback	✓
refund_amount	float	0.00	✓

- **Reconciliation:** SUM(psp.net_amount per settlement_date) ↔ bank credits — flag if delta > 2%

Synthetic Data Strategy

- Generate 200 synthetic merchants with Dutch e-commerce profiles (GMV EUR 300k–5M)
- Simulate 12 months of daily bank transactions per merchant
- Simulate PSP order-level data with realistic refund (2–10%) and chargeback (0.1–2%) rates
- Inject 5–10 'bad' merchants with deteriorating patterns
- Synthetic default labels via known formula — explicitly flagged as synthetic
- All generated via seeded numpy.random for reproducibility

Consent & Audit Logging

Every data access event is logged to DuckDB with: merchant_id, consent_type, scope, granted_at, expires_at, revocable flag. Business plan: 'borrower consent is logged and revocable.'

3. Feature Engineering Spec

The business plan specifies 80–120 interpretable features. Below are 15 core features for the MVP, fully specified with formulas.

Group A — Cashflow Stability

#	Feature	Formula	Missing Rule
1	monthly_net_cashflow_avg_6m	mean(monthly credits - debits) over 6m	Require ≥ 4 months; else DECLINE
2	revenue_volatility_90d	std(daily_rev) / mean(daily_rev) over 90d (CV)	Default 0.50 if < 60 days data
3	net_cashflow_coverage	monthly_net_cf_avg / (requested_amount / tenor)	Hard-knock if not computable
4	negative_balance_pct_90d	count(days balance < 0) / 90	Default 1.0 if bank data missing

Group B — Settlement & Payment Delays

#	Feature	Formula	Missing Rule
5	settlement_delay_p95	percentile95(settlement_date - order_date) days	Default 10 days
6	settlement_delay_median	median(settlement_date - order_date)	Default 5 days
7	supplier_pay_punctuality	count(on_time) / count(supplier_payments)	Default 0.70 if < 10 txns

Group C — Refunds & Chargebacks

#	Feature	Formula	Missing Rule
8	refund_rate_ltm	sum(refund_amount) / sum(gross_amount) 12m	Default 0.05 if < 6m PSP data
9	chargeback_rate_ltm	count(chargeback) / count(all_txns) 12m	Default 0.01
10	refund_trend_3m	refund_rate_last_3m - refund_rate_prior_3m	Default 0.0

Group D — Concentration & Seasonality

#	Feature	Formula	Missing Rule
11	platform_concentration	HHI = $\Sigma(share\_i^2)$ across PSPs	Default 1.0 (single platform)
12	seasonality_index	max(monthly_rev) / mean(monthly_rev) 12m	Default 2.0 if < 12m data

Group E — Ad Spend & Operational

#	Feature	Formula	Missing Rule
13	ad_spend_ratio_3m	sum(ad_debits) / sum(revenue) 3m	Default 0.20 if unclassifiable
14	bank_psp_recon_delta	abs(bank_credits - psp_net) / psp_net	Flag > 0.02 for fraud review
15	trading_months	(today - incorporation_date) / 30.44	From KvK registration

General Missing-Data Rules

- Bank data entirely missing → hard decline (NO_BANK_DATA)
- PSP data entirely missing → cap max amount at EUR 15,000, flag LIMITED_DATA
- Individual feature defaults are always pessimistic (worst plausible value)
- Features with > 40% missing across portfolio trigger DATA_QUALITY_ALERT

4. Code Build Plan

Directory	Purpose	Key Files
data/	Synthetic data generation & storage	synthetic_gen.py, *.parquet
ingestion/	Data loading, validation, reconciliation	base.py, bank_source.py, psp_source.py, reconciliation.py, consent_log.py
features/	Feature engineering pipeline	pipeline.py (15 features), missing_handler.py
policy/	Rules-based champion credit policy	credit_policy.py, knockouts.py, reason_codes.py
models/	Shadow ML training & scoring	train_shadow.py, score.py, model_card.md
decisioning/	Final decision assembly + audit log	decision_engine.py (DuckDB logging)
monitoring/	Drift, fairness, metrics	drift.py (PSI), metrics.py, fairness.py
app/	Streamlit demo UI	streamlit_app.py
notebooks/	Exploration & documentation	data_exploration, feature_analysis, shadow_eval

Repository: 35 files across 10 directories, fully packaged as ledger-mvp.zip

Quick Start:

```
pip install -r requirements.txt python -m data.synthetic_gen python run_pipeline.py streamlit
run app/streamlit_app.py
```

5. Key Code Modules

Module	Key Functions	Description
config.py	All constants	Thresholds, pricing bands, model params, DB path
data/synthetic_gen.py	generate_merchants(), generate_bank_transactions(), generate_psp_transactions()	200 merchants, 12m bank+PSP data, 5% bad merchants
ingestion/base.py	DataSource (ABC): load(), validate(), load_and_validate()	Abstract interface for all data sources
ingestion/bank_source.py	BankTransactionSource	PSD2 bank data loader with schema validation
ingestion/reconciliation.py	reconcile_bank_psp()	Cross-source fraud check: bank ↔ PSP delta
ingestion/consent_log.py	log_consent_event(), get_active_consents()	GDPR/PSD2 consent audit trail in DuckDB
features/pipeline.py	compute_features()	15 features from raw bank+PSP data per merchant
features/missing_handler.py	apply_defaults(), check_data_quality()	Pessimistic defaults + portfolio missing-rate alerts
policy/credit_policy.py	credit_policy()	Full champion: knockouts → scored gates → amount → pricing
models/train_shadow.py	train_shadow_model(), score_merchant()	GBM training (5-fold CV), pickle persistence, single scoring
decisioning/decision_engine.py	make_decision()	Combines policy + ML + explanations → DuckDB audit log
monitoring/drift.py	compute_psi(), run_drift_report()	Feature drift detection via PSI
monitoring/fairness.py	adverse_impact_by_sector()	Sector/platform approval rate analysis
run_pipeline.py	main()	End-to-end orchestrator: load → features → train → decide → report

6. Evaluation, Monitoring & Governance

MVP Metrics

Metric	Definition	Target (Year 1)
Approval rate	count(APPROVE + MANUAL_REVIEW) / count(all)	40–60%
Decline rate by reason	Distribution of reason codes among declines	Monitor; no target
Bad-rate proxy	% approved with synthetic default flag	Track only
Override rate	% credit-officer overrides vs policy recommendation	< 15%
Champion–Shadow agreement	% where policy decision and ML score > 0.5 agree	Track only
Avg ML score by band	Mean shadow score per pricing band	A < B < C expected

Stability & Drift Checks

- Population Stability Index (PSI) computed per feature
- PSI < 0.1 → stable | 0.1–0.2 → moderate shift | > 0.2 → significant drift (alert)
- Drift report generated weekly in production; on-demand in MVP

Fairness & Adverse-Impact

- No protected characteristics (gender, ethnicity, age) used as inputs
- Monitor approval rates by sector and n_psps for proxy discrimination
- Single-platform merchants may be disadvantaged by HHI — documented and reviewed
- Quarterly adverse-impact tables; no automated fairness adjustments in Phase 1

Model Documentation Artifacts

- **Decision log:** Every decision persisted in DuckDB with full feature vector, reason codes, ML score
- **Model card:** models/model_card.md — type, target, features, limitations, ethics, update policy
- **Consent log:** All data access events logged with timestamp, scope, revocability

Example Decision Explanation

```
LEDGER CREDIT DECISION — m_0042 ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ Decision: APPROVE
(pending credit officer review) Max amount: EUR 20,000 Pricing band: B (12.5% APR) Max tenor: 9
months Reason codes: * YEAR1_CAP — Year-1 pilot: max loan capped at EUR 25,000. Risk flags
(amber): * Revenue volatility (CV = 0.31) is moderate. * Platform concentration (HHI = 0.52) is
moderate. Shadow ML score: 0.08 (low risk — informational only) Next step: Credit officer review
required.
```

7. MVP Demo Script (5 Minutes)

Time	Step	What Happens
0:00	① Merchant connects accounts	Streamlit: 'Connect Mollie + bank' → mock consent logged
0:45	② Data ingestion	Bank + PSP data loaded → reconciliation check shown (✓/✗)
1:30	③ Feature computation	15 features displayed with green/amber/red traffic lights
2:15	④ Policy decision	Decision card: APPROVE/DECLINE/MANUAL_REVIEW + max amount + pricing
2:45	⑤ Explanation panel	Reason codes with plain-language explanations
3:30	⑥ Shadow ML comparison	Side-by-side: policy vs ML score. 'ML does not make the decision'
4:15	⑦ Risk dashboard	Portfolio: approval rate, decline reasons, drift (PSI), agreement rate

Key Talking Points

- **'We don't just look at the bank account'** — show how PSP data adds refund rate, settlement delays, and chargeback info
- **'Every decision is explainable'** — click through reason codes
- **'ML learns in the background'** — show shadow score, emphasize zero authority
- **'Governance from day one'** — show DuckDB decision log, model card, consent log

8. Final Checklist

■ What Is Real

Component	Status
Rules-based credit policy (champion)	✓ Real — fully specified, runnable
Feature engineering pipeline (15 features)	✓ Real — computes from raw data
Decision envelope with reason codes & explanations	✓ Real — production-shaped
Shadow ML training + scoring (GBM)	✓ Real code — trained on synthetic labels
DuckDB decision audit log	✓ Real — persists every decision
Reconciliation logic (bank ↔ PSP)	✓ Real — cross-check logic
Consent logging structure	✓ Real schema — mock data
PSI drift monitoring	✓ Real calculation
Repository structure + config	✓ Real — 35 files, buildable

■■ What Is Mocked

Component	Status
Merchant data (200 synthetic merchants)	■ Synthetic — no real merchants
Bank transactions	■ Synthetic — generated, not from AISP
PSP transactions	■ Synthetic — not from Mollie/Adyen API
Default labels (is_bad_synthetic)	■ Formula-based, not real repayments
KYC/KvK checks	■ Mock pass/fail — no real KvK API
Sanctions screening	■ Mock — no real PEP/sanctions database
Streamlit UI	■ Skeleton — not fully built yet

■ What Is Needed Before First Live Loan

Requirement	Category
Regulated AISP aggregator contract (live PSD2 access)	Data
PSP API agreement (Mollie and/or Adyen)	Data
KvK/UBO verification API integration	KYC
Sanctions screening provider	KYC
Head of Credit hired (credit policy sign-off)	Team
Legal review: lending license requirements	Regulatory
GDPR Data Protection Impact Assessment (DPIA)	Regulatory
Warehouse facility or equity-funded pilot budget	Funding
Collections workflow + arrears triggers documented	Operations
Real merchant onboarding (5–10 pilot merchants)	Validation
Replace synthetic labels with real 30+ DPD outcomes	Model
Credit officer UI for manual review workflow	Product
Penetration testing + encryption audit	Security